

Time Past and Time Present

With the core values of each culture's learning tradition having been outlined, I now turn to the central question of this book: Do these respective learning traditions still influence the present-day learners? In this chapter, I present empirical research that gives an affirmative response to the question. Research for the past several decades documents specific ways the present-day learners' beliefs and actual learning processes reflect their respective cultural learning traditions. However, before I present that empirical research, it is necessary to address some important questions about current education practice in China (and by extension, in East Asia as a whole) that may lurk in the minds of readers.

CHANGED CURRICULUM AND INCREASED COMPETITION

No one disputes that times have changed and that ancient values have undergone significant changes in much of the world. East Asia and the West are no exception. Yet, at least with regard to intellectual pursuit, the West has not undergone changes as radical as East Asia has, as noted in [Chapter 1](#). In fact, there is little regarding approaches to learning that the West has adopted from other cultures. Admittedly, educational content in the West – that is, what children learn about – has opened up to knowledge of other cultures and ethnic groups. Clearly, the most notable changes the West has witnessed are the broadening of compulsory education to all and the elevation of achievement standards for all.¹ Yet, we do not have evidence that any Western country has put in the core of their curriculum to teach children, for example, how to study as a Confucian or a Buddhist learner does. By contrast, education in China (and East Asia in general) has been fundamentally altered as a result of Western influence, ranging from its system (i.e., at what age children attend school, move to

the next level, etc.) to the very curricular content.² Thus, I focus more on the changes in China in this section.

Peculiarly, media and even some researchers have asserted that perhaps the present-day Chinese learners are no longer influenced by the Confucian learning tradition but have instead become more Westernized. I had a recent personal encounter that drives this perception home. I met a Mainland Chinese graduate student in sociology in Germany, who came to my research presentation on Chinese versus Western learning beliefs. During the break, we chatted. He asked me, “Do you really think that we have anything to do with Confucian values? I think that we are more Westernized.” I had a *deja vu* lapse. His question and remark were incredibly reminiscent of questions I asked during my undergraduate years. However, many Westerners with whom I have spoken view not only Mainland Chinese but all East Asians (including Asian immigrants with generations behind them in their host countries) as *very* different. Quite a bit of cross-cultural research demonstrates that Confucianism is largely responsible for that difference.

To be sure, educational curriculum, as noted previously, has changed in such a way that Western mathematics and science (and English at an increasing rate) have become the required core. The case can be made that learning fundamentally different subjects in school is bound to alter the learner’s approach to learning. After all, it is presumed that to succeed in learning math and science, learners must develop logical reasoning, causal inference, scientific methods, and objectivity in looking at the world. Furthermore, such learners need to adjust their purposes of learning as well. That Asian students have been documented to achieve well in these subjects adds more weight to the argument that East Asians must have changed or ought to have changed their approach to learning.

Moreover, the social conditions have changed dramatically, which increases demands on education. China has the largest population on earth, and since 1986, under their compulsory education law, every child must receive nine years of education.³ Population growth and compulsory education are similar in other Chinese and East Asian regions. These changes have resulted in a much greater need and pressure on the governments to educate people. Exacerbating this social burden is the fact that the world has been witnessing an explosion of knowledge, which also demands prolonged learning to ensure mastery. From the perspective of individual learners, competition for higher education, especially for the highly selective universities, is becoming more and more fierce as both the population and knowledge continue to grow. If desirable employment opportunities do not grow proportional to the highly educated workforce, the workforce may have to

retrain or upgrade skill sets. Thus, Chinese and other East Asian learners' learning beliefs and achievement may well be a direct response to these social changes, having nothing to do with Confucian learning tradition.

These arguments are well taken. However, with regard to the contention about learning Western subjects, the fact remains that before entering formal schooling, children have lived some five-to-six years under the intimate care of their families – in the Asian case, frequently of not just parents but the grandparents and extended kin. Although nuclear families are becoming the norm, they still remain more or less a living arrangement rather than a true nuclear structure as defined in the West. It is still very common for grandparents and even relatives to co-parent children; this pattern applies, surprisingly, also to Chinese immigrants who live in the United States.⁴ According to developmental research, the first few years are essential to children's later development.⁵ Therefore, during these very important formative years, Chinese children, as children from any culture, develop many important culturally informed beliefs and behavioral tendencies regarding learning,⁶ and they come with these beliefs and behavioral tendencies to the first day of school. As alluded to in [Chapter 2](#), the teacher–child relationship is modeled after the parent–child relationship in the Confucian tradition (more detailed empirical research on this topic will be discussed in later chapters). There is much more home-school coherence than may be the case in an ethnically diverse culture.⁷ Although children begin learning math and science, sometimes even during preschool, the social environment in which learning takes place is still fundamentally Confucian. Therefore, it is quite likely that while children learn math and science as subjects, the way they engage in learning may be different from their Western peers.⁸

With regard to social changes, Asia is not unique; most developed and developing nations also face similar educational challenges. However, nowhere is competition for education as intense as in East Asia. Pressure for children to achieve highly in school seems to be particularly pronounced. Again, this pattern also applies to East Asians living outside Asia in many societies where the mind, creativity, and intrinsic motivation are more prized than in East Asia. Moreover, although Western societies provide more educational opportunities and longer compulsory education than many Asian countries, competition for the best schools is just as fierce among Asian immigrants.⁹ Becoming the best in education seems to be the guidepost for Asians regardless of where they live. When asked “as a parent, what expectations do you have for your child's future?” one parent responded, “Magnet High, Harvard, and Heaven!” (Magnet High is a pseudonym for one of the

most competitive public high schools in the United States.) Desiree Qin, the researcher, describes this kind of desire for their children to go to the very best schools and universities as very common among immigrant Chinese parents whether they are from well-to-do or humble backgrounds.¹⁰

At this juncture, it is also important to acknowledge that popular press has portrayed Asian students' achievement as coming at a high price of their psychological well-being.¹¹ While comparable data are difficult to obtain from the Asian countries because of unclear reporting procedures and methods, available data do not support this view. A study comparing Chinese, Japanese, and American high school students' psychological adjustment in relation to parental and school pressure found that despite their reports of higher parental expectation and lower parental satisfaction with their achievement, Japanese students indicated fewer, not more, adjustment problems than their American peers. Chinese students also reported less stress, academic anxiety, and aggressive feelings than the Americans, although they did reveal higher frequencies of depressive mood and somatic complaints. Moreover, high achievement was not correlated with these adjustment problems for Asian students, but it was for American high achievers. Another recent study from Korea shows that parental pressure had positive, not negative, influence on adolescents' effort and achievement motivation.¹² The most recent data on suicidal thoughts among youths with major depressive episode (MDE, a strong predictor of suicide) from the U.S. Department of Health and Human Services show a lower rate of MDE among Asian youth than European-American youth aged twelve to seventeen.¹³ Finally, nationally representative data on Asian Americans' mental health indicate that as a whole, they suffer less, not more, mental problems than European-Americans.¹⁴ If parental pressure for school achievement truly causes these problems (because most Asian parents exert such pressure), Asian-American population as a whole should have a higher proportion of mental problems. The data, however, do not support this conclusion.

Recent research does indicate that Asian immigrant children in the United States display adjustment problems, particularly among adolescents. This issue merits our attention. However, academic pressure may be the easy scapegoat. If the Asian-American youth as a group do not show a higher rate of mental illness, and if some Asian immigrant children in the West show adjustment problems, then the *immigrant context* and the interactions between the home culture's childrearing values and practice and those of the mainstream culture may need to be studied in order to explain the true causes. Unfortunately, little research exists in this area. A few pioneering

studies shed light on some key factors, such as cross-language use (parents talk in their native language, but children respond in English, which leads to the breakdown in parent–child communication), the gap between desired parental warmth and the actual received warmth, decreasing time together, parents not understanding children’s changed worldviews, and children not understanding parents’ perspectives from their native culture all predict adjustment problems. But pressure to achieve has yet to be shown to be a direct cause of Asian immigrant children’s mental health problems.¹⁵

EXAMINATION HELL

The most astonishing part of the current Asian education system is its infamous examination system. That East Asian education system is examination-driven is no secret to the world. Any discussion of Chinese and Asian learning without acknowledging this condemned side is bound to face skepticism. This system has been accused of doing everything bad compared to the known good education practices from the West. The system, it is claimed, is notoriously teacher-centered and authoritarian, favors learning by rote, fosters extrinsic motivation, and stifles creativity.

As such, this system has been, quite unsurprisingly, under fierce criticism for quite some time (in fact, for more than a century). To express the aversion toward this system, Ishisada used the phrase “examination hell.”¹⁶ This loathing is echoed not only by non-Chinese researchers and educators, but also by many East Asian researchers and educators themselves. Nevertheless, anyone with some degree of sanity must ask the obvious question of why this notorious system still remains, is even flourishing today, and why Asians do not seem inclined to abandon it in spite of its having received such condemnation for so long. To be sure, many attempts to change the system have been made across East Asia (e.g., Japan and Taiwan). Yet, the system is no more likely to disappear than it ever was. Even more bizarre is the fact that most education policy makers in these societies are familiar with the “better” practices in the West, and many of these leaders were themselves trained in Western educational fields.

This fact is unsettling and demands some explanation. To my knowledge, no one has explained the legitimacy and vitality of this “examination hell” better than Samuel Peng.¹⁷ According to him, East Asian societies are “stuck” with the exam system because it is the only solution to a predicament with which East Asian cultures are faced. As stated previously, the Confucian tradition regards respect for and honoring of one’s family as the most important moral foundation for oneself. At the same time, Chinese

people and societies uphold the moral principle also espoused by Confucius: equality of education for all regardless of their backgrounds (有教無類). However, those holding power over educational access (e.g., college admissions officers) face a serious moral dilemma when their family members request favorable treatment (consider the case of a relative who just missed a few points on the college exam, which, in all likelihood, indicates only a trivial difference between students). If the request is denied, the person with institutional power violates the familial moral code; if the request is granted, the person violates public ethical standards. Alternative means of evaluating students, such as teacher recommendations and interviews, are all subject to the same dilemma. There is no other solution but to resort to the impartial test score. The only person responsible for advancement in education then is the student him- or herself. Viewed from this perspective, the examination hell may not be as hellish as it first seems.

This predicament would not exist in the first place if Chinese/East Asians as a whole, regardless of their socioeconomic status, did not desire and actually study to advance in learning. The real question here is why the Chinese/East Asians feel that they must pursue learning. Do they have nothing else meaningful in life to pursue other than academic learning?

A very well-established theoretical outlook among contemporary Western or Asian researchers alike is that school learning has one overwhelming purpose: providing an individual with knowledge and skills to make a good living and increasing opportunities for social mobility. Although researchers may not set out to study learning for social mobility, much of the actual research tends to focus on products of learning such as academic achievements and practical gains that reflect utilitarian purposes. Admittedly, economic survival goals may be valued by most, if not all, learners from any culture.¹⁸ However, if making a living were the only *raison d'être* for learning, we would not care about children's natural questions about the world, such as "why do maple leaves turn golden in the fall?" or "why is the sun bigger but colder in the morning but smaller yet hotter at noon?" Neither would we admire adolescents who engage in philosophical debates about, for example, the meaning of justice. Nor would we continue to read Socrates' demonstration of human innate knowledge in his tutelage of an uneducated slave boy, or Confucius' words about commitment to self-perfection in learning. But we do, and we do so diligently.

Making a living may be the convenient reason for pursuing academic learning among people at lower socioeconomic status. It does not, however, explain learning behaviors of affluent people. I happen to teach at a university where there is a mixture of affluent and low-income but very bright

European-American, as well as quite a few Chinese/East Asian and Asian-American students from both affluent and humble backgrounds. It is my observation that regardless of their socioeconomic status, the European-American students engage in learning with similar attitudes and behaviors – that is, emphasizing personal curiosity, inquiry, verbal communication, and individual insights and brilliance. Likewise, it is also common to see many Chinese/East Asian students from affluent homes study just as hard as their peers from low-income backgrounds. These students do not need to make a living by studying hard; their families can ensure them a comfortable life. But they go through the ordeal of hard work willingly or feel they need to honor their parents by dedicating themselves to study even if they would not take on the ordeal so willingly for themselves. Chinese immigrant learners throw this issue in an even sharper light. These learners enjoy freedom, choice, and creative environments available to them at U.S. schools. Yet, they consistently display the same degree of seriousness and achievement as their peers in East Asia, which cannot be attributed to the pressures of the examination system. Thus, we are compelled to think that learning for the present-day European-Americans and Chinese/East Asians has to serve other purposes as well. As it turns out, the compelling forces behind their respective learning approaches are still their own cultural traditions.

EXPLAINING ASIAN STUDENTS' ACHIEVEMENT IN EARLIER RESEARCH

There was virtually no research on comparative learning beliefs or the related area of achievement motivation between Western and East Asian learners prior to 1960. In 1963, David McClelland published a paper on the motivational pattern of Chinese, in which he declared that traditional Chinese lacked achievement motivation, but thanks to the communist ideological spirit, Chinese people gained revolutionary zeal and became more motivated.¹⁹ After this publication, there was a long silence on the subject until the late 1970s, when international assessment of educational achievement revealed that Japanese students outperformed their counterparts in many Western countries in math and science, with the United States falling behind other developed countries.²⁰ This rather unexpected outcome suddenly spawned an interest in understanding East Asian schools and children's learning. Herald Stevenson and James Stigler launched a decade-long landmark research that compared American (mostly European-American) and Asian elementary school students from Japan, Taiwan, and China. Their central objective was to find out the causes of Asian children's higher

academic achievement. The researchers overcame great challenges to conduct this multinational, multilingual research. They ensured that elementary school students were representative in their own countries. They constructed their own achievement tests in math to control for bias, collected surveys from children and parents on their views of achievement, observed classroom teaching, and interviewed teachers.²¹

The results were quite startling. They found that whereas American children, parents, and teachers explained children's achievement based on the notion of ability, their Asian counterparts attributed children's achievement to their effort. They dubbed this learning difference the "learning gap." They also found this major distinction of beliefs to be linked to how schools and the school day were organized. American children attended school for fewer days and shorter time daily than Asian children. Asian teachers assigned more homework, and parents expressed dissatisfaction with their children's achievement even when the achievement was already very high. Teachers were not reluctant to display and correct children's mistakes publicly and even to make children's achievement public in the classroom (i.e., everyone knew how everyone else did on any test). American teachers assigned less homework, and parents showed more satisfaction with their children even when the achievement was mediocre. American teachers would not do what Asians teachers did to expose children's mistakes and weaknesses in public. Quite the contrary, they would keep children's achievement confidential to protect their self-esteem.²²

This research occupied the center stage and generated much more research interest. For example, American researchers studied American and Japanese concepts that learners used to describe their ability and effort, and their analysis yielded that Japanese learners display culturally specific dispositions such as *seishin*, the mental attitude that helps one tackle a task, and *gambaru*, a "positive orientation toward the intrinsic benefits of ... persistence."²³ Researchers in Hong Kong took the concept of *effort* to study Chinese students. Effort had been defined, in Western motivation literature, as an internal but unstable factor (meaning that a person may or may not exert effort depending on the achievement task). They found that effort was not situationally unstable but, in fact, a stable factor across time and situations for the Chinese learners. In other words, Chinese students believed that they need to make an effort all the time for all learning tasks.²⁴

Others also focused on U.S. and Asian curricular and classroom differences within school and found that Japanese schools provided more opportunities for children to learn math and that their curriculum was more differentiated and better sequenced in intensity according to the

advancement of grades.²⁵ Joseph Tobin, David Wu, and Dana Davidson observed and videotaped the routines of a preschool in the United States, Japan, and China for one day. They showed the representative footages to the teachers, administrators, and parents of each school as well as those of the other two participating schools. Their video-ethnography captured details of how children's daily activities were organized and guided, how they were instructed in class, and how they interacted with peers. This research revealed that the U.S. preschool education emphasized children's self-expression, free exploration, enjoyment, smooth peer interaction, and enforcement of social and moral rules of school. Chinese preschool stressed nutritional care, collective activities under teachers' close supervision, disciplined learning in class, and following social and moral rules strictly. Japanese preschool exhibited yet another feature: fostering children's learning among themselves. Teachers were mostly hands-off, leaving children to play, fight, and resolve their conflicts on their own. Quite surprisingly, there was little academic learning.²⁶ Similarly, Catherine Lewis documented Japanese elementary school teaching and learning in an in-depth ethnography. She further showed that Japanese schools organized instructions and activities in a way that engaged not only children's minds but also children's affect in learning.²⁷ Finally, Robert Hess and Hiroshi Azuma studied teaching styles in classrooms of U.S. and Japanese young children and discovered that whereas U.S. teachers preferred an efficient, well-paced style of instruction, their Japanese counterparts used a slow but thorough style. These different styles predicted children's working habits and achievement in their respective cultures.²⁸ More researchers also documented effective teaching strategies in math and science in Japanese elementary classrooms, which promoted student achievement.²⁹ The best-known effort was, however, the video-study led by James Stigler to compare classroom teaching in the United States, Japan, and Germany. The study concluded that Japanese pedagogy indeed had advantages compared to its Western counterparts.³⁰

Research attention was also paid to the familial factors. Researchers investigated European-American and East Asian parents and found that although parents in both cultures valued education highly, they differed in several areas. Asian parents expressed higher expectations of their children's achievement (usually straight A's compared to A's and B's by European-American parents). Asian parents also engaged their children in more structured learning activities such as enrolling them in afterschool or weekend enrichment programs. Asian parents also monitored their children's home learning more closely than European-American parents did.³¹

Meanwhile, achievement in math and science by East Asian students from Singapore, Korea, Japan, Hong Kong, and Taiwan continues to rank at the top of the global rankings for the past three decades, as again confirmed by the more recently published results from the Trends in International Mathematics and Science Study (TIMSS) and the Programme for International Student Assessment (PISA) results. It seems that regardless of how the students are tested, they remain at the top. Furthermore, they show larger improvements even in reading (e.g., Korea ranked sixth and Japan eighth in PISA, 2003). Asian immigrant students in North America, Europe, Australia, New Zealand, and other parts of the world also consistently show higher achievement. Note that all of these TIMSS and PISA assessments did not involve China until 2009 (when China's Shanghai participated in PISA) probably because of China's (then) lagging economic development. Given their first placement in their only participation in one 1990 international assessment, they were unsurprisingly placed in 2009 PISA as first with 17 points for reading, 54 points for math, and 37 points for science above the second-place Korea, followed by Finland and other East Asian countries who differed by only a few points (all these scores are based on a mean score of about 500).³²

THE PARADOXICAL CHINESE LEARNER

Against this international backdrop, somewhat detached from U.S. interest in East Asian learning, a group of researchers from Australia, the United Kingdom, and Sweden working in Hong Kong wrestled with a phenomenon that they termed the paradox of the Chinese learner. The basic problem is that whenever Westerners teach Chinese students or visit Chinese schools, they are disappointed with, if not outraged by and condemnatory toward, Chinese learning. As noted previously, the Chinese educational system consists largely of old-fashioned teaching and learning. On the one hand, pedagogy is teacher-centered, authoritarian, with a centralized curriculum (implying inflexibility and lack of attention to individual children's learning needs). On the other hand, students are docile, obedient, and uncritical; they learn by rote, lack intrinsic motivation, and aim only for exams. In other words, everything known about good educational practice is violated in the Chinese/Asian educational system. I also remember attending a graduate course at Harvard Graduate School of Education. When I tried to share with the instructor that practicing writing Chinese characters or calligraphy really helps the child learn how to concentrate, the instructor dismissively said "All Chinese students do is rote learning! That is not what

we are dealing with in this course.” These remarks shut me up for the rest of the course. Indeed, at that time, I did not even know how to respond. It was clear to me, however, that Chinese learning was regarded as a negative model. As it turned out, my personal experience was not unique. There are plenty of articles, books, and popular press that have criticized East Asian learning.³³

Yet, time and again – and by now it is quite unambiguous – Chinese/Asian learners achieve well in comparison with students from other developed countries. They continue to do well when they come to the West for advanced studies, and more, their immigrant children in the West are also highly achieving. All these findings indicate to me that Chinese/East Asian children achieve well no matter how they have been (often quite exhaustively) assessed. This stark contrast between poor learning approaches and high achievement has led keen observers of Chinese learning such as John Biggs and David Watkins to regard the whole phenomenon as begging for a new explanation. Indeed, how could such an educational system with such learners produce any meaningful achievement?

Watkins and Biggs assembled a team of researchers who contributed to their influential volume, *The Chinese Learner*, in which they posited the phenomenon as a paradox.³⁴ Because most East Asian learners are under the influence of the Confucian learning tradition, Watkins and Biggs used the term *Confucian heritage cultures* (CHC) to include countries of China, Hong Kong, Taiwan, Singapore, Japan, Korea, and Vietnam. The contributors presented empirical research that documented basic beliefs and learning processes observed in learners that are distinctly CHC. In 2001, Watkins and Biggs published a sequel of their first book, *Teaching of the Chinese Learner*.³⁵ Despite the focus on teaching, most research was conducted in close relation to learning. Alongside these volumes, other researchers also collected telling data that added weight to the needed explanation of the paradox. In the next section, I present the research findings from these efforts.

Learning Beliefs

With regard to cultural beliefs, Wing On Lee provided a historical account of core Confucian values concerning learning and argued for their enduring impact on Chinese learners today. Accordingly, most essential is the Confucian belief in human self-perfection pursued as the highest purpose of life through personal commitment to learning. This belief is also linked to other beliefs such as one’s social contributions in the form of

meritorious service and practical concerns for honoring their families as well as enhancing their own social status and mobility.³⁶ Lee's argument is coherent with the Confucian learning tradition reviewed in [Chapter 2](#). In support of Lee's argument, Kai-ming Cheng collected data on people's beliefs about learning in a comprehensive ethnographic study on one Chinese province's primary education. He concluded that Chinese parents, whether well off or destitute, send their children to school not to learn literacy and numeracy skills, but to become a person who is knowledgeable of the world, able to function well in social relations, and, most important of all, morally cultivated.³⁷ Similarly, An Ran's qualitative research examined how Chinese parents in Britain and British teachers clashed on the purpose of learning. Whereas British teachers focused on acknowledging and expressing satisfaction with Chinese children's apparent high achievement, Chinese parents were discontented, emphasizing more demanding learning materials and their children's continuous effort to self-improve regardless of their accomplishments.³⁸

Related to these studies, Lixian Jin and Marin Cortazzi found that the image of a good teacher as described by British students is one who is able to arouse students' interest, explain clearly, use effective instructional methods, and organize activities. However, the image of a good teacher offered by their Chinese peers is one who has deep knowledge, is able to answer questions, and is a good moral model.³⁹ Similarly, Lingbiao Gao and David Watkins used both qualitative and quantitative methods to identify different goals of teaching held by Chinese science teachers. They emphasized cultivating students' adaptive attitudes toward learning and moral guidance in addition to other cognitive goals.⁴⁰ Irene Ho interviewed Hong Kong and Australian teachers and found that Australian teachers viewed their teaching within the framework of professional responsibility with clearly defined roles and boundaries. As such, they would notify the families if students failed or misbehaved, but they did not see as their responsibility to teach morals to their students. Instead, the family was more responsible for moral instruction. In comparison, Hong Kong teachers stressed their moral charge of guiding students on the "right path and were prepared to go all the way to rectify misbehaviour, spending as much time as they could and often getting personally involved." They attributed students' academic failure and misbehaviors more to their own inadequacy than to students themselves or their families.⁴¹ Finally, Thomas Tang used qualitative methods to study Hong Kong teachers' conceptions of learning and teaching and found that moral development is the highest learning as well as highest teaching conception beyond the more cognitively oriented conceptions.⁴²

In the studies where comparative perspectives were taken, Chinese students and teachers, but not their Western counterparts, emphasized moral and personal growth as a central purpose of learning and teaching.

Learning Processes

With regard to the learning process itself, Ference Marton, Gloria Dall'Alba, and Lai Kun Tse tackled rote learning and memorization, the notoriously Chinese/Asian style of learning that has received much criticism from the West as well as from Chinese educators themselves. However, it turned out that Chinese rote learning was not an end in itself but was used as the first step of a larger strategy for achieving deeper understanding.⁴³ In a related study comparing British and Chinese students' use of memorization and repetition, Bo Dahlin and David Watkins further found significant cultural differences. Whereas British students used repetition to check if they really remembered something, Chinese students used it to create "deep impressions" to lay a foundation for developing understanding. Moreover, British students viewed understanding as a process of sudden insight, whereas Chinese students believed understanding to be a long process that required extensive mental effort.⁴⁴

In exploring perceptions of effective teaching by Hong Kong college students and faculty (both Chinese and Western expatriate), Daniel Pratt, Mavis Kelly, and Winnie Wong found that Chinese students and faculty held different views of learning from those of Western teachers. Western teachers often characterized Chinese students "as not knowing how to think, having only short term goals ... wanting to be spoon-fed, needing too much structure ... taking a quiet, receptive, and deferential attitude during class, and [unwillingness to] challenge or question ... authority."⁴⁵ However, Chinese students believed that learning is a gradual process that requires tremendous dedication and methodical steps (similar to Japanese learning and teaching style as documented by Robert Hess and Hiroshi Azuma).⁴⁶ Generally, they engage in four distinct steps to accomplish any learning task. Upon encountering new materials, Chinese students initially commit the material to memory; next they seek to understand the intention, style, and meaning of the material. They then try to apply their understanding to situations that call for use of such knowledge, and finally they enter a deeper level of questioning and modification of the original material. Whereas the last step in their approach is verbally interactive by nature, the first three steps may call for more solitary learning and contemplation (which is an important aspect of Chinese intellectual tradition⁴⁷). Clearly,

this style is not bound by the immediate verbal exchange at the moment but can extend over a period of days, weeks, months, and in some cases even several years (as a doctoral student may publish a paper to challenge his or her mentor's ideas with which the student disagreed several years earlier)! Thus, these researchers observed that Chinese students often feel frustrated and bewildered when they are confronted with a Western teacher "whose expectations and forms of assessment thrust them immediately to the far end of this chain (questioning and analysis)."⁴⁸

With regard to Asian learners being driven by extrinsic instead of the more desirable intrinsic motivation, one study shed some important light on this topic: Sheena Iyengar and Mark Lepper examined how personal autonomy and choice functioned among Asian-American and European-American schoolchildren in learning and achievement situations. Personal autonomy and choice are hallmarks of intrinsic motivation, whereas social influence and determination (such as choice made by others) are forms of extrinsic motivation. It has long been assumed that personal autonomy and choice are conducive to learning and performance, whereas the lack of such personal freedom is detrimental.⁴⁹ However, these researchers demonstrated that this assumption held true for European-American children but not for their Asian-American counterparts. The former enjoyed learning and performed better when given personal choices (of what to learn and how to learn it), whereas the latter enjoyed the learning just as much and did better when their task was chosen by significant others (e.g., mothers or trusted peers). More recent research further documents that Asian children learn well and enjoy learning activities chosen by others with whom they have good relationships. Thus, Asian learners' need for personal choice may not play as essential a role as for Western learners.⁵⁰

FURTHER RESEARCH: DIGGING INTO CULTURAL LEARNING BELIEFS

More than two decades of research has greatly enhanced our understanding of Chinese/East Asian learners. Notably, most of recent research has been done with impressive qualitative methods designed to uncover valid cultural meanings behind the observed differences in learning processes and outcomes. Although small in size, Hong Kong offers a unique research ground for such truly cross-cultural research because of its unique free-trade status. Researchers in Hong Kong from both CHC and Western cultures interact with each other frequently, and more importantly, many Western teachers

interact with CHC learners and schools directly. This research has indeed charted new grounds for further research.

Mapping Cultural Learning Models

Although learning research by the 1990s had made significant advancement, the accumulated knowledge as a whole remained quite discrete and lacked a systematic approach to learning from a *cultural* perspective. There was no comprehensive description of learners in the West and Asia, or, for that matter, in any other culture. The separate pieces of information may have represented fragmented learner images, but this information explained neither learners' culturally based beliefs nor the underlying reasons and processes that produce such learner beliefs. If our goal was to understand human learning from all perspectives, a comprehensive documentation of cultural learning models – the missing link – was in order.

I began researching this topic by adopting what is called an *emic* perspective. An emic perspective is the insider perspective, originating with the people being studied, as opposed to the one originating with the outside observer/researcher, which is called an *etic* perspective.⁵¹ Scholars have advocated for the inclusion of both emic and etic views in any research on human cultures (or else cross-cultural understanding would be impossible). However, traditional research has been dominated by the etic perspective, particularly in the form of theory-driven and hypothesis-testing research modeled after the Western scientific paradigm. Anthropological research was the avant-garde in introducing the emic perspective and has helped us achieve a great deal of understanding of the world's cultures. Research from the emic perspective tends to have high validity and is preferable to the etic perspective in uncovering native concepts and the beliefs that underlie observed behaviors of people.

I also explored this research topic from the framework of *cultural models*, again an anthropological insight. Cultural models are established by historical processes of the culture and continue to be revised as the culture evolves further. They are conceptual frames that shape members' experiences by supplying the group's shared ways of explaining, predicting, and interpreting people's thoughts, feelings, and behaviors. These frames also guide people in forming their goals and motivate them toward obtaining their goals.⁵² An example of a cultural model is the image of a teacher in the United States. Anyone who has gone through school from K to 12 in the United States probably knows what a teacher does, what power and responsibilities he or she has, what students ought to do in relation to the teacher,

and what happens when a student fails to do what the teacher is charged to demand from the student. Similarly, Western and Chinese learning beliefs can be studied as such cultural models.

I conducted two studies comparing Chinese and European-American learning models. The specific empirical methods I used are called *prototype* methods. These methods are based on the theory and empirical evidence that much of human understanding of the world consists of categorizations of objects, activities, and experienced events. For example, furniture as a higher-order abstract category contains functional categories of sofas, tables, chairs, and so forth, which people use for daily life. With each subcategory, there are finely differentiated types such as couch, bench, and stool for people to sit on. Humans achieve this categorization system with the aid of language by giving objects names and labels. Once these categorical structures exist, they become prototypes of objects, models, or scripts of human activities and events, according to which people categorize new objects, activities, and events. For example, when we see a chair-like object, we are likely to use the prototypical chair to conceptualize the object and subsequently categorize it as chair.⁵³ Thus, the language of a given culture has these prototypes for people to use and to organize their understanding of the world. Prototype methods lent themselves to the study of learning as conceptualized by humans living in specific cultures.

To access the language that contains learning conceptions in each culture, we asked three native-speaking students from selective universities in each culture to free-associate the Chinese term *xuexi* (學習) and *learn/learning* in English, respectively.⁵⁴ To ensure that the Chinese and English terms were equivalent in meaning, several steps were taken. First, we consulted the word frequency dictionaries in both Chinese and English. Word frequency is a standardized linguistic index for how frequently a word is used in a natural language. The higher the frequency, the faster and more accurately the word is recognized by the users of that language.⁵⁵ We chose the Chinese term *xuexi* (*learn/learning*) because it occupies a word frequency of 679, the highest of all the Chinese learning synonyms. Similarly, we identified *learn/learning* because its word frequency of 254 was the highest among its synonyms.⁵⁶ Then, we asked twenty fluent bilingual adults, with half native-speakers and the other half non-native-speakers who had acquired high fluency with the other language, from each culture to translate the Chinese term into English and also translate the English term into Chinese. The cross-translation and a rating procedure (see more discussion later) led to the conclusion that *xuexi* and *learn/learning* were the closest equivalents among synonyms in both languages.⁵⁷

The three college students in each culture respectively wrote down common words and phrases denoting aspects of learning that came to mind when they heard the word *xuexi* and *learn/learning*. This step resulted in an initial list of 242 in English and 145 in Chinese. Next, we presented each list to twenty more college students from similar backgrounds in each culture and asked them to add any items they thought referred to learning. This step was necessary to cast the net wide so that important cultural learning concepts were not left out. We obtained an expanded list of 496 in English and 478 in Chinese. Finally, it was quite likely that these lists contained items that might not have been shared by most cultural members, but instead represented idiosyncratic ideas of individuals. To obtain a core list likely shared by most cultural members, we asked sixty more participants in each culture to rate each item on a four-point scale according to its relation to learning with, “1” meaning no relation and “4” a definite relation to learning.

Following research conventions, we used the median number 2.72 as a cut-off line in order to select a core list from all the rated items.⁵⁸ We carefully examined each item and selected a final core list of 203 English and 225 Chinese items, representing a reasonable degree of relevance to learning as determined by group consensus. Table 3.1 shows the top English and Chinese terms referring to learning.

As can be seen in Table 3.1, there are striking differences in these two lists of learning terms. Table 3.2 summarizes four types of differences immediately observable by glancing at Table 3.1. With regard to purely linguistic features, English terms are mostly single and regular words. By contrast, Chinese terms typically have multiple words with many modifiers. Many terms are idiomatic expressions in the form of proverbs and sayings. Regarding the conceptual features, English terms have references to external factors such as resources, institutions, and teacher. The highest-rated items on the Chinese list have no such references (although there were such references with lower-rated items). Interestingly, among the nearly 500 initial English terms, there were no terms referring to hard work. In sharp contrast, many Chinese terms refer to hard work and related learning attitudes (this finding does support the research by Stevenson and Stigler as reviewed earlier). Neither does the English list contain terms on life-long learning, but the Chinese list has several terms including the very top item (“keep on learning as long as you live”). There is one category that has the reversed trend: The English list has many terms denoting thinking and mental processes (61 terms, or 30% of the entire list of 203 terms), but only one such term appears on the Chinese list of the top 20 terms (and only 14

TABLE 3.1. *Top twenty learning-related terms nominated and rated by U.S. and Chinese adults*

English	English Translation (Chinese)
1. Study	Keep on learning as long as you live (life-long learning) (活到老, 學到老)
2. Thinking	Read extensively (博覽群書)
3. Teaching	Learn assiduously (刻苦學習)
4. School	Read books (看書)
5. Education	Diligent (in one's learning) (勤奮(學習))
6. Reading	Extensive knowledge and multifaceted ability (博學多才)
7. Teacher	Study (讀書)
8. Books	Make a firm resolution to study (發奮讀書)
9. Critical thinking	Study as if thirsting or hungering (如飢似渴地學習)
10. Brain	There is no boundary to learning (學無止境)
11. Discovery	Concentrate on learning (專心學習)
12. Understand	Eager to learn (好學)
13. Information	Take great pains to study (苦心攻讀)
14. Knowledge	Seek knowledge (求知)
15. Motivation	The learned understands reasoning (讀書人明理)
16. Library	Study abroad (留學)
17. Students	Do one's utmost to self-study (勤勉自學)
18. Learn by doing	Learning without thinking is labor lost; thinking without learning is perilous (Confucius) (學而不思則罔, 思而不 學則殆 [孔子])
19. Applying ideas	After learning, one understands that one's knowledge is inadequate (Confucius) (學然后知不足 [孔子])
20. Communication	Long-term diligence is the road to the mount of knowledge; endurance of hardship is the boat to the boundless sea of learning (書山有路勤為徑, 學海無涯苦作)

terms overall, or 6% of the entire list of 225 terms, with the remainder having no reference to either logic or analysis).

With respect to affective features, the English terms lack affect. The Chinese terms, in contrast, express strong affect, showing desire, passion, and intensity. Finally, in reference to behavioral features, there is no clear call for action in the English terms. The Chinese terms exert a strong call for action.

The significance of these noted differences in the learning vocabulary becomes quite clear when we consider the process by which children develop their learning beliefs in these two cultures. In [Chapter 7](#), I present data on how the two cultures' mothers converse with their children about learning. Mothers use many of the words and phrases captured in my initial

TABLE 3.2. *Linguistic, conceptual, affective, and behavioral differences in English and Chinese learning lexicon*

English	Chinese
Linguistic features	
Single words	Multiple words with many modifiers
Regular words	Idiomatic expressions including proverbs and sayings
Conceptual features	
Reference to external factors (e.g., resources and institutions)	No such reference
No references to hard work	Many references to hard work and learning attitudes
No references to life-long learning	Several such references
Many references to thinking/ mental processes	Only one such reference
Affective features	
Affectively neutral	Strong affect (desire/passion/intensity)
Behavioral features	
No clear call for action	Clear call for action

studies to guide their children in their discussions of their offspring's good and poor learning attitudes and behavior. Children develop their own learning beliefs by hearing and using these terms. Thus, the learning vocabulary plays an important role in child development.

To map out the relationships among various conceptions for each culture, we asked 100 college students from similar backgrounds to sort their culture's core list into groups according to similarity in meaning. These sortings were then submitted to the statistical analysis called *cluster analysis*, which resulted in two conceptual maps of learning, one for English and another for Chinese as shown in [Figures 3.1](#) and [3.2](#).⁵⁹

Both maps contain much detailed information, showing the magnitude and complexity of each culture's learning model. There are two dimensions on each map: levels and clusters. Levels refer to the hierarchical structure of categorization as discussed previously in the section on prototype methods. As can be seen, both maps consist of four levels. At the very top is the Superordinate Level, below it is the Basic Level with two further levels, called Basic Level I and Basic Level II. Further down is the Subordinate Level, which has the most divisions. It is this level that contains the actual learning terms. The clusters show similarities and differences in learning conceptions as sorted by the participants. They are found at the Subordinate Level that

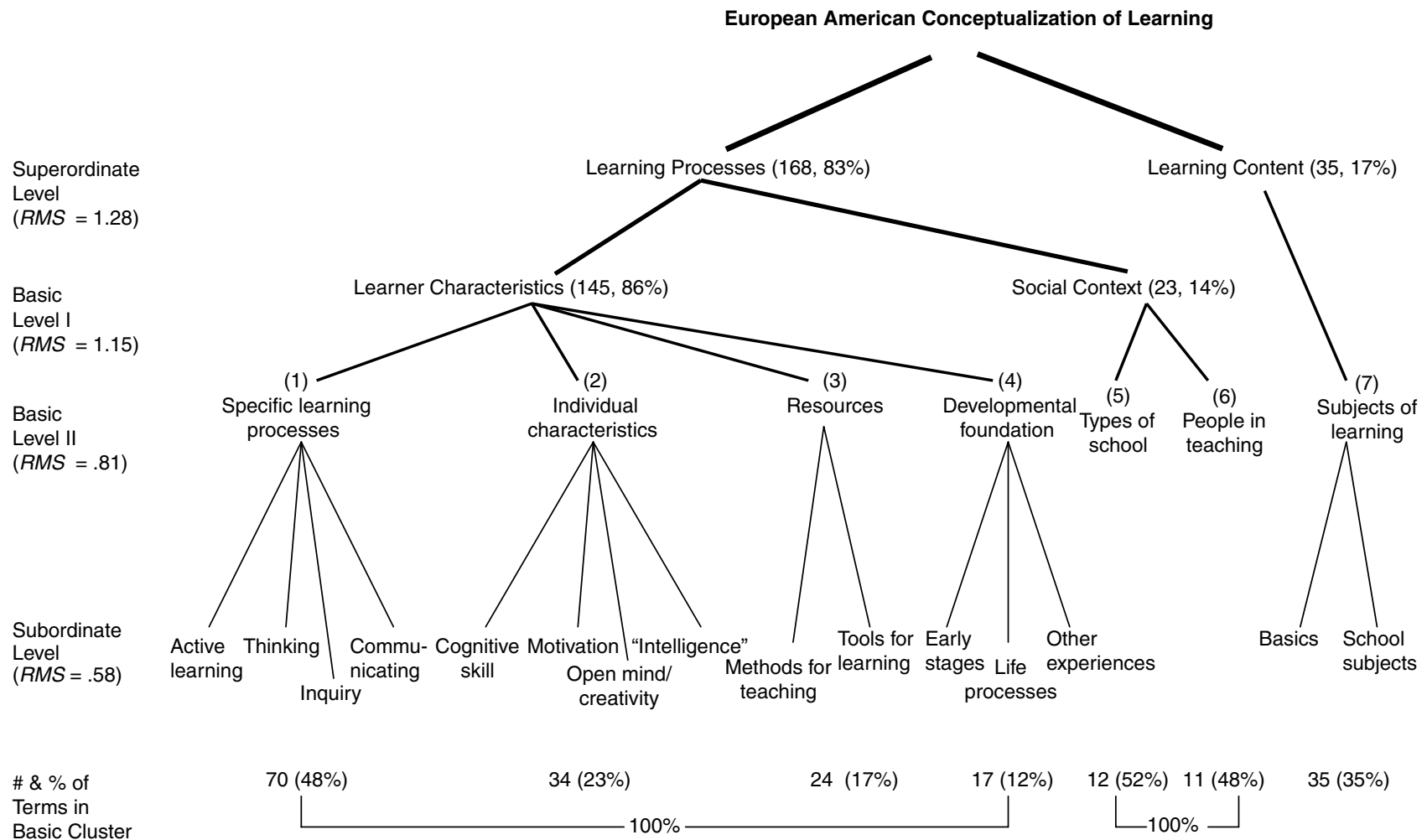


FIGURE 3.1. Diagram for a European-American hierarchical structure. From J. Li (2003). "European-American conceptualization of learning," *Journal of Educational Psychology*, 95, p. 261. Copyright 2003 by the American Psychological Association. Adapted with permission.

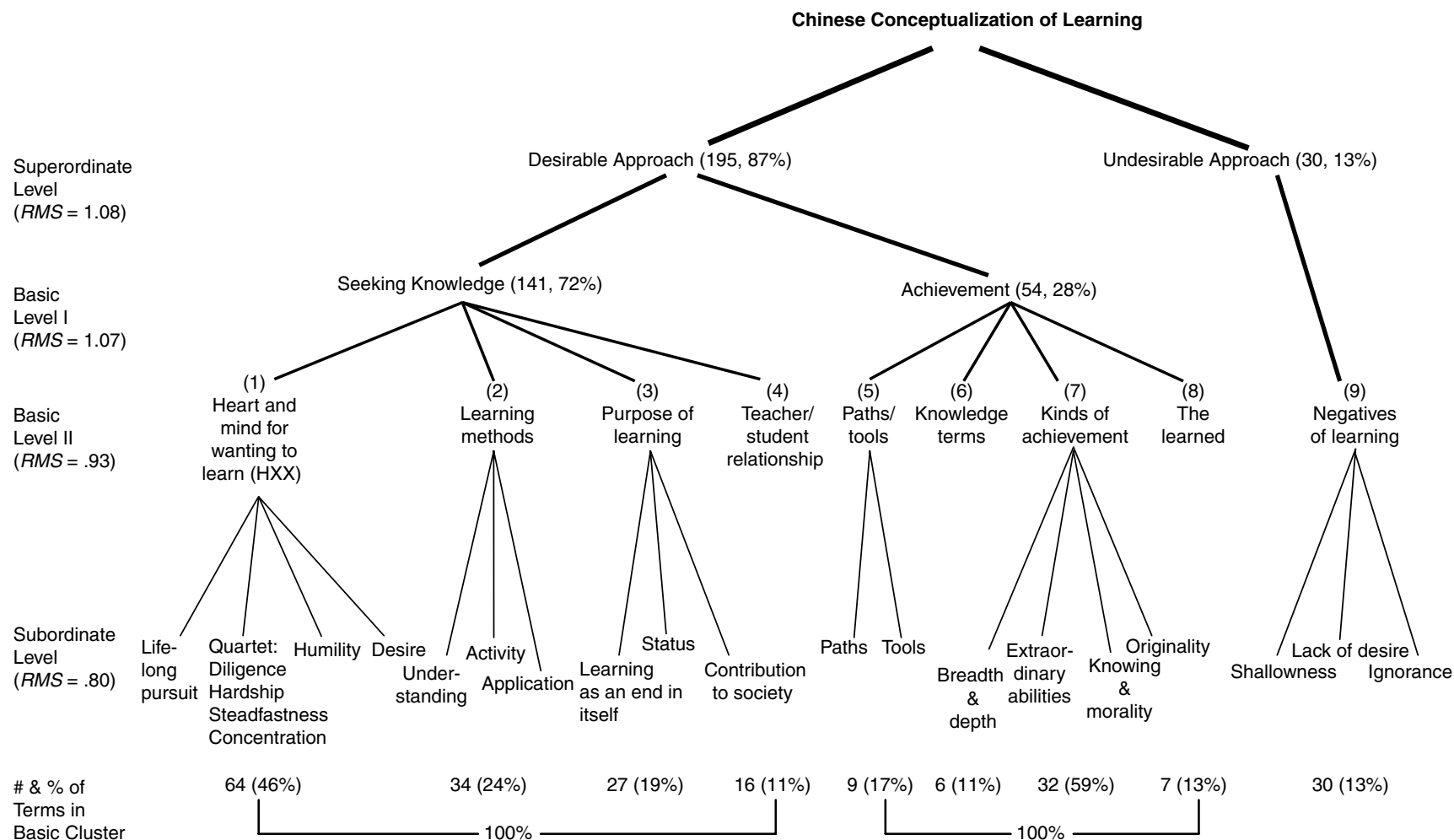


FIGURE 3.2. Diagram for a Chinese hierarchical structure. From J. Li (2001). "Chinese conceptualization of learning," *Ethos*, 29, p. 123. Copyright 2001 by the American Anthropological Association. Adapted with permission.

literally “clusters” words and phrases from the lexicons into the various divisions. How many clusters there are and how large each cluster is are determined by how many people sorted how many learning concepts into various groups. Based on the assumption of prototype methods, a greater number of items indicate greater awareness/emphasis of the conceptions in that culture.⁶⁰ Because these maps are empirically derived with group consensus, they are the *culture-level* learning beliefs, which are taken to represent a form of cultural learning models as accessed through language usage.

For our present purpose, it suffices to highlight the most relevant features. The U.S. map (Figure 3.1) focuses on learning processes (with the majority of terms) on one side and learning content (with fewer terms) on the other. Within the learning processes, a great many more terms fall within learner characteristics than within social context. The two most significant dimensions are under learner characteristics: (1) specific learning processes elaborating on (a) thinking, (b) inquiry, (c) active learning, and (d) communicating; and (2) individual characteristics stressing (a) cognitive skills, (b) motivation, (c) open mind, and (d) intelligence.

The Chinese map (Figure 3.2) displays desirable versus undesirable approaches to learning with a preference/value at the top. The majority of terms fall on the desirable side, which contains two further distinctions: seeking knowledge and achievement standards. Under seeking knowledge, the most significant groups are: (1) heart and mind for wanting to learn (in Chinese, *hao-xue-xin*, 好學心), which includes (a) life-long pursuit, (b) a set of learning virtues (diligence, endurance of hardship, steadfast perseverance, and concentration), (c) humility, and (d) desire; and (2) purpose of learning, containing three essential ideas: (a) learning as an end in itself, (b) status, and (c) contributions to society. Under achievement there is one significant dimension: kinds of achievement emphasizing breadth and depth of knowledge, abilities, unity of knowing and morality, and originality.

In the second study, we asked 62 college students for each culture to describe their ideal learners in their respective cultures (124 in the sample).⁶¹ We targeted ideal-learner images instead of average learners because they provide what developmental researchers term the “optimum” of developmental outcome. Accordingly, cultures have preferable “endpoints” or “optimal ways of being” toward which younger members of a culture are enculturated.⁶² Although no two children will achieve identical outcomes in the enculturation process because of diversity in individual characteristics and contextual influence, it is important to describe the ideals because they exist in people’s minds and guide people’s behavior.⁶³ The ideal-learner image, as opposed to words and phrases in the first study, would also

compensate for the limitations inherent in the lexicon from which I derived culture-level learning beliefs. Ideal images are thus also culture-level, but not individual-level, learning models.

We probed four specific dimensions of the ideal learner: (1) thinking on the nature of knowledge, purposes and processes of learning, and views of intelligence and excellence; (2) understanding of the relationship between learning and one's moral development; (3) learning behaviors in routine situations any learner is likely to encounter, such as high achievement, high intelligence, failure, not understanding concepts, inability to learn despite effort, and boredom; and (4) emotional patterns associated with good or poor learning. The written descriptions of each of these dimensions were analyzed both qualitatively and quantitatively.⁶⁴ These procedures yielded four profiles corresponding to the four probed dimensions of the ideal learner for each culture.

The basic findings from the two studies converge to two comprehensive pictures of the two cultures' learning models. Table 3.3 summarizes the components and dimensions of these two belief systems. As can be seen, there are at least four large common component headings across the two cultures: purpose, agentic process, affect, and achievement. The specific items within each component were determined by the number of each culture's respondents who referred to these ideas as well as the presence of these components on the two cultural maps of learning concepts as derived from the first study of learning terms. Below I discuss each culture's model while highlighting similarities and differences between the European-American and Chinese models.

European-American Learning Model

Purpose of Learning

As indicated in Table 3.3, three main purposes emerged: (1) cultivate the mind/understand the world, (2) develop one's ability/skill, and (3) reach personal goals. The most frequently mentioned associations with the idea of the purpose of learning are to cultivate one's mind and to understand the world. The mind enables one to learn, but it also develops or becomes sharpened as a result of exposure to a stimulating environment that demands its proper use for understanding the world.

The second learning purpose is to develop the ability/skills that one needs to be a fully functioning member of one's society. Important skills include those for a successful career, for one's self-sufficiency and independence, as well as knowledge that enables one to solve problems and to

TABLE 3.3. *Components and dimensions of European-American and Chinese learning models*

EA	Chinese
Purpose of Learning	
Cultivate mind/understand world	Perfect self morally/socially
Develop ability/skill	Acquire knowledge/skills for self
Reach personal goals	Contribute to society
Agentic Process of Learning	
Active engagement	Diligence
Thinking	Self-exertion
Inquiry	Endurance of hardship
Communication	Perseverance
	Concentration
Kinds of Achievement	
Understanding of essentials/expertise	Breadth-depth/mastery of knowledge
Personal insights/creativity	Application of knowledge
Being the best one can be	Unity of knowledge and moral character
Affect	
Positive	
Curiosity/interest	Commitment (“establish one’s will”)
Intrinsic enjoyment	Love/passion/thirst
Challenging attitudes	Respect/receptivity
Pride for achievement	Humility for achievement
Negative	
Indifference/boredom	Lack of desire
Extrinsic motivation	Arrogance
Disappointment/low self-esteem for failure	Shame/guilt for failure

Source: Li, J. (2011). Cultural frames of children’s learning beliefs. In L. Arnett Jensen (Ed.), *Bridging cultural and developmental approaches to psychology: New syntheses in theory, research, and policy* (pp. 26–48). New York: Oxford University Press; p. 30. Copyright 2011 by Oxford University Press. Reprint with permission. Originally adapted from table 14–1 in Li, J. & Fischer, K. W. (2004). Thoughts and emotions in American and Chinese cultural beliefs about learning. In D. Y. Dai & R. Sternberg (Eds.), *Motivation, emotion, and cognition: Integrative perspectives on intellectual functioning and development* (pp. 385–418). Mahwah, NJ: Erlbaum.

take control of one’s life and surroundings. The third learning purpose is to reach one’s personal goals (beyond ability/skill). This purpose includes finding self-fulfillment, achieving personal happiness, becoming a well-rounded person, embarking on a spiritual quest, or reaching any personal goals individuals may desire.

These three types of purposes are inherently related. While the mind plays a central role in learning, it also needs cultivation. This very process is

also one by which a person develops various abilities and skills in order to reach personal goals in life.

Agentic Process of Learning

The features under this heading are called *agentic* because they are personal actions directed at achieving desirable goals. In other words, these are the processes in which people engage in order to learn, given their purposes and goals.⁶⁵ Four main agentic processes were found. First is active engagement, centering on the notion that a person needs to be actively involved in learning. This active nature emphasizes learning by doing and life experiences, both inside and outside the classroom. The second process is captured in the idea of “thinking/critical reasoning,” which is what the mind does or ought to do. Thinking concerns the whole spectrum of mental processes involved in learning. For example, one could move from lower-order differentiations of objects to higher-order synthesis of relations, or one could engage in rigorous deductive logic or inductive reasoning, or one is free to reflect, introspect, or contemplate on anything of personal interest.

Inquiry, the quintessential process and agency underlying Western scientific development, is the third kind that stresses discovering the unknown and inventing the new. In this process, one seeks to find out about things in the world through a variety of research activities such as gathering data, experimentation, observation, analysis, and drawing conclusions. Key to this process is one’s engagement in challenging existing canons of thought, finding new problems, and searching for creative solutions.

Finally, the fourth process, communication, serves both as learning itself and a form of dissemination of one’s knowledge and discoveries. For learning itself, one participates in oral as well as written forms of social interactions. In these communications, one not only exchanges ideas with others, but also discusses, debates, critiques, or argues with others in order to achieve better understanding of a subject, using tools such as logic, evidence, and devil’s advocacy. For dissemination purposes, one presents, explains, articulates, and demonstrates one’s views, positions, or research findings, often using visual forms and technical devices to reach audiences and readers.

Like the purposes, these four agentic processes in learning are also linked coherently, even though they each have distinct emphases. Active learning ensures fuller engagement of the mind, which promotes thinking at all levels, forms, and dimensions in inquiry. Communication is a natural part of active learning given its interactive nature, which in turn facilitates more active engagement, thinking, and inquiry.

Kinds of Achievement

Three kinds of achievement were most frequently identified (Table 3.3): (1) understanding of essentials/expertise, (2) personal insights/creative problem solving, and (3) being the best one can be. Understanding of essentials is not limited to a particular field of study but includes anything deemed worthy of learning by the person. Expertise is a notion that applies only to an area of study, such as math or art. Given that the purpose of learning is to understand the world, the standard for gauging achievement is sensibly the extent to which a person gains better understanding of a subject. Many respondents asserted that excellence of learning does not rest just on knowing facts but on knowing the deeper underlying principles of things – that is, the essence of the phenomenon. Personal insight and creative problem solving, the second standard, displays one's brilliance and creativity. The third kind of achievement addresses being the best one can be in learning. Whereas the first two standards imply less social recognition, the third standard is a more subjective and intrapersonal delineation of "being one's best." One sets his or her own goal of achievement and strives for it.

Of these three kinds of achievement, the first two are more closely related with one another than with the third one, although when striving to be one's best is aligned with understanding the world, developing expertise in a field, and applying one's knowledge/skill creatively to solving problems, the three kinds are synergistically linked and inform each other.

Affect

Affect is an integral part of learning beliefs and processes. Whenever human beings pursue their goals with any degree of effort, they experience related emotions and feelings. Whereas emotions typically involve physiological responses such as heartbeat, muscle tension, facial expressions, and bodily gestures, feelings may not involve such.⁶⁶ For example, if we feel strange in a situation, we may not experience bodily change. Nevertheless, our feeling strange is part of our affective system that allows us to detect noteworthy changes in our environment.⁶⁷ Affect in our research refers to all emotional responses as well as longer-term feelings toward learning and self-engagement in the learning process as has been defined and used in the research on human motivation.⁶⁸

We found both positive and negative affects in our research on European-American learning model (Table 3.3): For positive affects there were (1) curiosity, interest, and motivation for learning, (2) intrinsic enjoyment, (3) challenging attitudes, and (4) pride in achievement. For negative affects

there were (1) indifference and boredom, (2) extrinsic motivation, and (3) disappointment/low self-esteem for failure.

Curiosity, interest, and motivation were the most often mentioned positive affect for purposes. Intrinsic enjoyment was named as the affect that accompanies the learning process. Attitudes that challenged existing canons of knowledge and authority were greatly prized. The general positive affect associated with achievement was pride – an expression for self-confidence and self-esteem. These positive feelings about oneself in turn motivated the person to learn further.

Negative affects occur when the learning task is perceived as uninteresting or boring. When this happens, ideal learners were described usually as not giving up, but continuing to learn the materials, especially if the materials were still required or were important knowledge. Still, they would not display intrinsic motivation and enjoyment. Instead, they would persist with some level of indifference, boredom, even dread. Many respondents also acknowledged that these learners would stop investing time and effort into the materials if they were deemed less essential. Relatedly, respondents referred to extrinsic motivation as antithetical to natural curiosity, interest, and enjoyment.

When experiencing failure, ideal learners were described to feel a number of related negative emotions: disappointment, lack of confidence, low self-esteem, and inferiority. Naturally, these emotions stand in sharp contrast to those associated with high achievement.

European-American Learning Model Echoes Western Intellectual Tradition

Taken together, it is remarkable that the findings regarding purposes and agency of learning reflect well the four key dimensions of the long-standing Western intellectual tradition since Greek antiquity as discussed in [Chapter 2](#). The purpose of learning remains to understand the external world. Although this external world has expanded to the social world, human psychology, the fictional world, the brain, and even the self, the approach to these worlds is to treat them as *objects of analysis* rather than as a process of personal transformation by the learner him- or herself. Mind still remains supreme as a human capacity to crack the worlds, and the process of obtaining verified, objective, and reliable knowledge is still prized. A critical attitude toward existing knowledge and inquiry into all phenomena is still essential in any course of learning and research. The achievement standards are by and large

framed also within the Western intellectual tradition. Individual learners' achievement is judged by how well they understand their chosen world of study. Their personal excellence is intimately tied to the ultimate purposes and goals on the one hand and the learning processes on the other.

Similarly, their affects mirror the fascination, wonder, and intrinsic passion about the world as well as the inquisitive and critical spirit that characterize Western scientists since the Greeks first displayed such affective tendencies. Although historical materials typically do not elaborate on affective responses to failure by the Greeks and subsequent scientists, the findings of our research on negative affects about personal failures were sensible, given the emphasis on those positive affects.

Chinese Learning Model

Purpose of Learning

Similar to the European-American model, there are also three main purposes: (1) perfect oneself morally/socially, (2) acquire knowledge/skills for self, and (3) contribute to society (Table 3.3). The first, the most significant one, was described as a need to perfect oneself in the moral and social realm. It is important to point out that some European-American respondents also mentioned self-fulfillment and self-actualization as their ideal learner's purposes. However, they emphasized being one's best through learning academic subjects rather than cultivating themselves morally as defined in Confucian terms. Interestingly, the term *learning* was left open for respondents to construe. Whereas European-Americans interpreted it as understanding the world, their Chinese peers took it to connote moral and social self-perfection in addition to academic learning.⁶⁹

The second purpose, acquiring knowledge/skills for self, is reminiscent of the European-American second purpose, developing one's ability and skill. Chinese respondents stressed mastery of knowledge whereas European-Americans emphasized developing ability. Chinese ideal learners also needed knowledge/skill for leading a good life. They, too, must have those skills for survival, self-sufficiency, and successful careers. Finally, these skills were seen as enabling and empowering them to solve problems, maintain satisfying social relations, and reach their personal goals.

The third purpose is contributing to society, which reflects directly the core Confucian value. As Rosemont suggests, Chinese children are beneficiaries of their social world's love, care, and teaching, but they gradually become benefactors as they gain knowledge, skill, and moral maturity.⁷⁰

Individual learning is not divorced from contributing to community as an ultimate purpose.⁷¹

These Chinese purposes are also inherently related. They have been explicitly part of Confucian values regarding learning and are actively promoted by families, communities, schools, and society at large.⁷² Therefore, one needs to engage in personal skill learning and moral development before one can meaningfully contribute to society. However, one's moral self-perfection and skill acquisition are not conceptualized as separate or sequential processes. They take place simultaneously from early on and continue throughout life. Many European-American ideal learners also desired to make a difference in the world, relieve suffering, and help others. However, this purpose was not as strongly or prevalently expressed as by their Chinese peers.

Agentic Process of Learning

When asked to describe how their model learners learn (i.e., what they do, what steps they take, how they behave when they encounter difficulties such as failure or boredom), Chinese respondents did not, surprisingly, describe much that would be considered learning processes per se as seen in European-American descriptions. Instead, they wrote extensively about the learning virtues, as discussed in [Chapter 2](#). These learning virtues also appeared in my study on Chinese learning terms (see “quartet” in [Figure 3.2](#)).

Five such virtues emerged. The first, the notion of earnestness/sincerity (*renzhen*, 認真/誠意), a variation of the ancient spirit of sincerity in learning (as discussed in [Chapter 2](#)), stresses the seriousness that one needs to bring to learning. The seriousness is sensible given that Chinese learning is centered around one's moral self-perfection. As respondents described, earnestness is the predecessor of dedication, which is the long-lasting virtue all learners are urged to develop.

The second virtue, diligence/self-exertion (*qinfen*, 勤奮/發奮), refers to frequent studying behavior after the person has made a resolve to pursue learning. The emphasis falls on the actual learning behavior, therefore much time spent on learning, following the person's resolve.⁷³ *Qinfen* as a behavioral manifestation of self-exertion is believed necessary to ensure one's clarification of goals; the course of action one must take to realize them; and a way to hold oneself accountable for any temptation to stray from one's resolve. Frequently, upon making the resolve, the person shares his or her learning goal with his or her family, close friends, or teachers,

who serve as witnesses to monitor, to watch for, and even to demand consistency between one's resolve and follow-up action.

The third virtue, endurance of hardship (*keku*, 刻苦), focuses on overcoming difficulties and obstacles one is bound to encounter in learning. Respondents described three kinds of difficulties: (1) physical drudgery and poverty, (2) difficult knowledge, and (3) lack of natural ability. First, physical drudgery and poverty are considered hardships because they have been an unavoidable living condition throughout Chinese history (see [Chapter 2](#) for an account of this virtue as part of Chinese learning tradition). Even though living standards have improved in recent decades, physical labor and poverty remain the harsh reality for many. Difficulty in understanding particular academic subjects or concepts is a routine hurdle for any learner. Finally, respondents were very clear about individual differences in their natural capacity and acknowledged the associated impact on people's learning. However, there was also consensus that these obstacles are not reasons for not learning. Instead, ideal learners developed endurance of hardship, which would enable them to face and combat these particular obstacles.

The fourth virtue, perseverance (*hengxin*, 恆心), addresses a general attitude toward learning and a behavioral tendency in learning. Perseverance is valued because of the belief that there is no shortcut to learning. Knowledge does not come about overnight, but through a bit-by-bit, accumulative process over a long period of time, a process fraught with obstacles and distractions.⁷⁴ Perseverance is believed potent in helping one stay on the task from the beginning to the very end. It is a virtue required to achieve any serious learning.⁷⁵

The final virtue, concentration (*zhuanxin*, 專心), is used in Chinese more often to describe a general learning behavior, not necessarily related to specific tasks. Concentration emphasizes studying with consistent focus and unswerving dedication. It also includes carefulness and thoroughness of learning.⁷⁶ Concentration is believed to be an essential quality of the learner because this disposition allows the full engagement of one's mind and heart in study.⁷⁷

These five learning virtues are clearly related and form a whole. They all presume a desire to learn because without it, these learning processes and behaviors cannot be sustained. Without resolve and its resultant commitment, diligence, endurance of hardship, and persistence may be limited to sheer situational factors. Likewise, if concentration can be halted by "hardship" or if one lacks perseverance, one's resolve may be aborted halfway.

Chinese respondents also revealed other learning processes and activities similar to those revealed by their European-American peers, such as thinking, observation, participation in social activities, and talking to people. However, compared to their European-American peers, they mentioned those categories significantly less. Similarly, European-American respondents also acknowledged their ideal-learner's hard work and persistence, but their reference to these learning virtues was less central and consistent.

Kinds of Achievement

There were also three general kinds of achievement similar to the European-American achievement (Table 3.3), but the two sets of achievement standards convey very different meanings. The first one, depth and breadth and/or mastery of knowledge, is also captured in Figure 3.2 under "kinds of achievement." Whereas breadth refers to one's extensive knowledge of different disciplines or subjects, depth concerns one's profound understanding of one subject. Moreover, the integration of breadth and depth is also emphasized. Mastery may not highlight breadth and depth, but it nevertheless stresses possession of knowledge, and by implication, the broader and deeper such possession, the better. Mastery together with breadth/depth is sensible considering that the ultimate goal is self-perfection, which is open-ended and life-long in nature.

The second standard, application of knowledge, is reminiscent of the European-American personal problem solving. However, the Chinese emphasis falls on the *use* of what one has learned in real-life situations. The conceptual distinction lies in book knowledge versus knowledge in use. Whether such use is personal or social in origin matters less. This standard thus includes applications that may not be deemed as creative or insightful in any sense (e.g., use math to verify a bank transaction). The third standard is unity of knowledge and moral character, also shown as one kind of achievement in Figure 3.2. Consistent with the purposes of moral and social self-perfection, acquisition of knowledge for self, and contribution to society, achieving the unity of the cognitive/intellectual with the social/moral makes the elements compatible and harmonious.

These achievement standards are inherently related to each other. Acquisition of depth and breadth of knowledge can enable the person to better apply such knowledge, which in turn can broaden and deepen one's knowledge. So long as the learner also continues to self-perfect morally and socially, he or she will likely continue to seek breadth and depth of learning, which loops back to his or her ability to use his or her knowledge in life.

Affect

Chinese affect also contained both positive and negative types. Four positive affects emerged: (1) commitment (“establish one’s will”), (2) love, passion, and thirst for learning, (3) respect and receptivity, and (4) humility for achievement. There were three negative affects: (1) lack of desire, (2) arrogance, and (3) shame/guilt for poor learning.

The notion of commitment, establishing one’s will (*lizhi*, 立志), is part of one’s learning purpose. This concept aims at helping the learner, often during secondary school, start pondering his or her life’s purposes in order to come to a clear personal vision (*zhixiang*, 志向 or *baofu*, 抱負). In doing so, Chinese learners believe that they will not only find a more specific path to focus on (e.g., I want to be an engineer), but also know to what path to attach their energy and dedication (e.g., therefore I will study math, physics, and engineering). Therefore, the process of *lizhi* is a spiritually very uplifting and emotionally positive process.

Some European-American respondents also touched on personal ambitions. However, such cases were not described consistently as a deliberate and socially concerted process where the learner is urged to search for a purpose and to establish commitment to reaching it.

Love, passion, and thirst were described similarly as enjoyment among European-American ideal learners. However, a significant difference lies in the source of such affect. Whereas for the European-American learners, intrinsic enjoyment, curiosity, and interest were described as essential, this intrinsic source was not emphasized by Chinese respondents. In fact, many acknowledged that their ideal learners were initially not motivated or interested in learning at a young age, but they developed love and passion once they realized the importance of learning, or once their parents and teachers guided them into the process. This kind of formation of love and passion as cultivated and fostered by one’s social world parallels the recent work on intrinsic motivation as less essential for Asian-American children for school learning.⁷⁸

Respect/receptivity is another distinct affect that Chinese ideal learners express toward knowledge and teachers.⁷⁹ Because learning in the Confucian persuasion is not limited to academic learning but also includes social and moral learning, respect toward knowledge and teachers (that ideally embody the self-perfecting process) is sensible and expected. However, this general attitude of respect/receptivity among Asian learners has been taken as a sign of docility, obedience, and lack of critical thinking.⁸⁰ This is a misunderstanding. Asian learners’ deference toward knowledge and teachers does not stem from their fear or blind acceptance of authority, but

from their deep sense of humility. Instead of treating humility as a personal weakness, they regard it as personal strength and courage, because those who are humble are willing to self-examine, admit their inadequacies, and self-improve. Therefore, respect/receptivity and humility go hand in hand. When experiencing feelings of inadequacy, one's ego or self-esteem is not seriously threatened and in need of protection as may be the case among European-American learners.⁸¹ Chinese learners believe that one can always self-improve so long as one learns humbly and respectfully from others.⁸²

As alluded to earlier, Chinese learners' receptivity and humility may be very different from European-American learners' challenging attitude, especially in the form of immediate verbal exchange in the midst of a class or a discussion. It was generally the case that Chinese respondents made few references to such challenging attitudes toward teachers, even though their ideal learners did engage in debates with their peers. However, this does not mean that Chinese learners do not challenge. In fact, many respondents wrote that challenging old knowledge or advancing new knowledge was an important goal for learners in the end. But one is reluctant to engage in challenges until one has thoroughly understood the knowledge in question or mastered one's field.⁸³

For achievement, Chinese learners generally display less pride even if they may be happy themselves.⁸⁴ This tendency is different from their European-American peers who usually feel proud of themselves and like to share their joy with others.⁸⁵ Chinese ideal learners were described as feeling a need to remain calm and humble. Relatedly, the need to be humble comes from the same recognition that learning is a life-long journey. Although others may acknowledge one's achievement publicly, focusing on celebration for oneself may be perceived as a negative tendency that pulls one away from further self-perfection.

The two negative affects are lack of desire and arrogance. The former shows some affinity to the European-American indifference and boredom. Lack of desire is the opposite of a heart and mind for wanting to learn.⁸⁶ Many Chinese learning-related terms refer to this state as being at the heart of any motivational problem. By using or hearing these words, learners sense disapproval and concern from their social world. Arrogance is also an affect that is the opposite of humility. Learners who achieve highly are believed to be particularly vulnerable to this inflated sense of self. Given the importance of humility in the Chinese learning model, there is little wonder why pride/arrogance is a great concern among learners.

Similar to their EA peers, Chinese-model learners were described as feeling a number of related negative emotions such as sadness and pain when

they experience poor learning. The most frequently revealed emotions are shame and guilt both for themselves as well as their families.⁸⁷ Shame is a powerful and prevalent emotion in Chinese culture and is an emotion of disgrace or humiliation as in most cultures. However, it is also a moral discretion and sensibility that people desire to develop.⁸⁸ Thus, the meanings of shame and guilt shade into each other. Together they function to direct people into self-examination in order to recognize their own wrongdoings as well as to motivate people to mend and to improve themselves. Given that learning is geared toward self-perfection for Chinese lives, feeling shame/guilt in poor learning is to be expected.

Chinese Model and Its Confucian Lineage

As alluded to throughout the preceding section, the key components as found in the research also reflect well the four key themes of the Chinese learning tradition as outlined in [Chapter 2](#). Perfecting self, as the most important purpose of learning, continues to inspire Chinese learners as a whole. Although few college students in my study used the expression “taking the world upon oneself (以天下為己任), they unequivocally named contribution to society as a major purpose of their learning. This latter expression is the modern manifestation of the ancient “take the world upon oneself.” They convey the same meaning by linking individuals’ learning to a larger social and therefore moral purpose beyond personal benefits from learning.

With regard to learning virtues, the empirical findings virtually match the traditional phrases and expressions. In fact, learning virtues may be the most intact and potent survival of this Chinese learning tradition. Perhaps this is because learning virtues are more an individual psychological matter, and their cultivation and maintenance are not readily subject to political changes. The process of forming such learning virtues takes place via family socialization and teacher–student interactions modeled after parent–child relationships, rather than in receiving or embracing political ideologies. So long as the political and large social movements do not vehemently attack these learning virtues, they have little reason to change (e.g., consider the possibility that the emancipation of women from homes to schools and workplace may have intensified the transmission of these learning virtues instead of destroying them – there are now twice as many people developing and exercising the learning virtues in school as both genders are schooled).

Finally, none of the learning virtues and purposes of learning emphasize self-expression and verbal communication. Chinese learners appear to think and behave in accordance with the principle of “action is better than words.” They prefer showing by doing over speaking about what they know.⁸⁹

CULTURE-LEVEL VERSUS INDIVIDUAL-LEVEL BELIEFS

The aforementioned research that I conducted provides descriptions of the European-American and Chinese learning models at the cultural level, which is not to be taken as a model of individuals. These learning models are considered as cultural models for two reasons. First, the models are derived from learning-related lexicons that are the properties of the two cultures, the languages that belong to the cultures, not to specific individuals. Second, the derived ideal-learner images are not specific real persons, but rather two generalized ideal images, despite the fact that a sample of participants, taken as representative of each culture’s learners, described their ideal image.

Each cultural learning model is recognizable by the members of the culture. It also serves to influence and guide (but not determine) the members’ thinking and action in learning. Individual learning beliefs are not termed as models because they are specific to individuals. Their own beliefs may differ from their cultural model as a function of social categories such as gender, socioeconomic status, and region of residence, by context (e.g., political turmoil and increased educational opportunities), as well as by other changing factors such as age and role (e.g., parent versus student). Although cultural models do evolve over time, they tend to be stable relative to individual-level beliefs, which can change more rapidly as a result of changed circumstances or development of the person.

NOTES

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- or Westernization? In L.-H. Fan, N.-Y. Wong, J.-F. Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 27–64). Singapore: World Scientific for a comprehensive review of the similarity in current East Asian and Western mathematics curriculum.
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 4. See Chen, F.-N. (2005). Residential patterns of parents and their married children in contemporary China: A life course approach. *Population Research & Policy Review*, 24(2), 125–148 for research on how grandparents still care for their grandchildren and married children still care for their ailing parents. See Li, J., Holloway, S. D., Bempechat, J., & Loh, E. (2008). Building and using a social network: Nurture for low-income Chinese American adolescents' learning. In H. Yoshikawa & N. Way (Eds.), *Beyond families and schools: How broader social contexts shape the adjustment of children and youth in immigrant families* (pp. 7–25). New Directions in Child and Adolescent Development Series. R. W. Larson & L. A. Jensen (Series Eds.). San Francisco, CA: Jossey-Bass for research on extended family's co-parenting of Chinese American children.
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of the Organization for Economic Co-operation and Development (OECD) on immigrant children from and to different countries. Among the four sending countries – Turkey, former Soviet Union, ex-Yugoslavia, and China – the first three showed large variability with regard to where the students came from, where they landed, what language they spoke at home, and what SES their families had. However, Chinese immigrant children achieved highest irrespective of these factors. That Chinese children in Macau, Hong Kong, New Zealand, and Australia continue to achieve well cannot be attributed to the particular regions they came from, their host countries, schools they attended there, and their SES levels. The head of OECD, Andreas Schleicher remarked: “[I]n general, socio-economic status has less impact in East Asian countries than in Western European ones.” If these key factors did not impact Chinese learners, then we are compelled to look somewhere else to explain their learning. I suggest that the way in which Chinese parents socialize their children and the way Chinese learners engage in learning may be crucial. See also Ran, A. (2001). Traveling on parallel tracks: Chinese parents and English teachers. *Educational Research*, 43, 311–328 for clashes between immigrant Chinese parents and British teachers’ expectations of children’s learning. See Zhou, M., & Kim, S. S. (2006). Community forces, social capital, and educational achievement: The case of supplementary education in the Chinese and Korean immigrant communities. *Harvard Educational Review*, 76, 1–29 for manifestations of Confucian learning values among Chinese and Korean immigrant families in the United States.

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87. See the first reference in note 61 to this chapter.
88. See the second and third references in note 51 to Chapter 2; and Fung, H., & Chen, E. C.-H. (2001). Across time and beyond skin: Self and transgression in the everyday socialization of shame among Taiwanese preschool children. *Social Development*, 10(3), 420–437.
89. See Chapter 8 for a presentation of the most recent research on speaking versus action among EA and Chinese people.